

AASD 4016

Full Stack Data Science Systems

Applied AI Solutions Developer Program



Module 5

Deployment

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Agenda



- Deployment
- Dockerhub
- Creating GCP account
- GCP instance
- SSH
- Installing Docker
- Running the docker container
- Testing the public endpoints

Iris Classifier

Notebook to Script

Script to Flask

Load model and keep it up for inference

Infer from the model

Test the endpoint using Curl, POSTMAN

Build a python web client

Test the client

Create a Virtual Environment

Install all libraries

Create requirements.txt

Create a Dockerfile

Create a Docker image

Create a Docker container

Test the endpoints within the running

Docker container

**Push the docker image into
Dockerhub**

Create GCP Compute Engine

SSH into GCP instance

Install docker

**Pull the docker image from
Dockerhub**

Run the docker container

Test the public endpoints

Current Flask project structure

Project Structure

```
app.py
helper(s).py [Optional]
templates/
    index.html
    result.html
static/
    Style.css [Optional]
```

Dockerfile

requirements.txt

iris-venv/

.dockerignore

Deploying a web application

Installing Docker

Creating a Virtual environment

Installing necessary libraries

Creating requirements.txt

Creating a Dockerfile

Building a Docker image

Running a Docker container

Testing the endpoints

**Push docker image into
Dockerhub**

Creating GCP instance

SSH into instance

Install docker

Pulling a Docker image

Running a Docker container

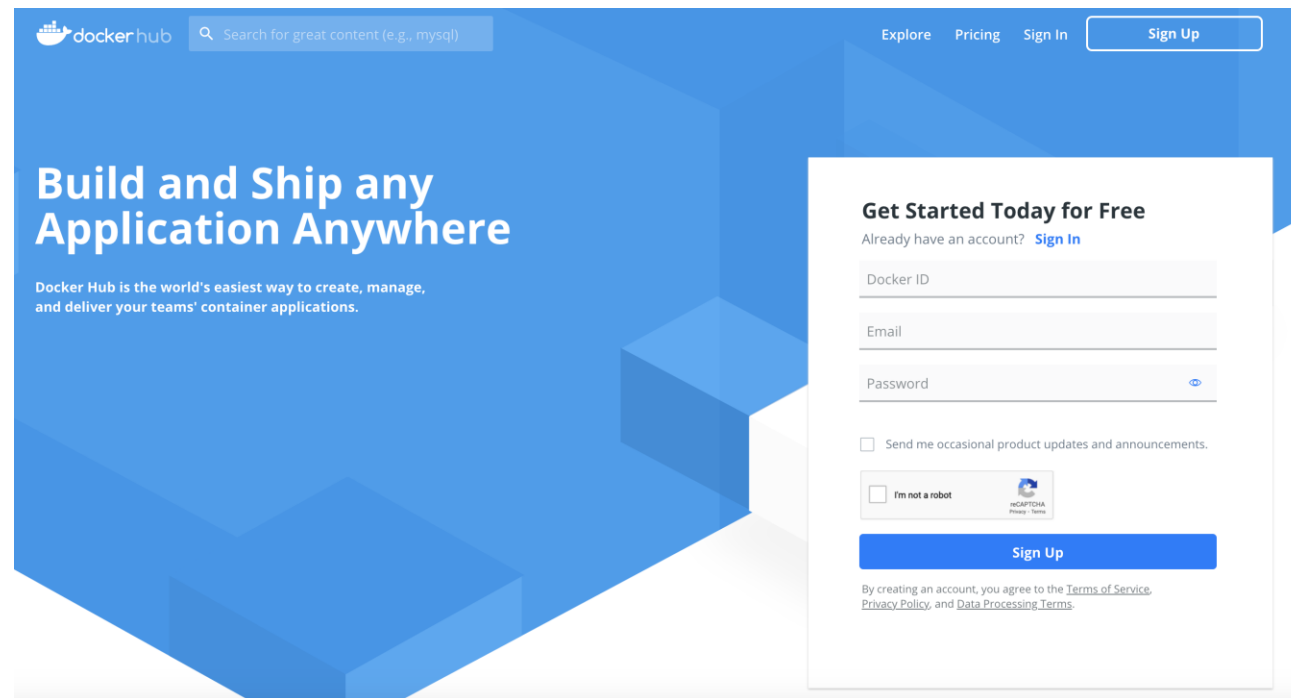
Testing the public endpoints

Dockerhub

Dockerhub

Create a Dockerhub account

`https://dockerhub.com`



The screenshot shows the Docker Hub website's sign-up interface. The background is a blue geometric pattern with the text "Build and Ship any Application Anywhere" and a description of Docker Hub. A white sign-up form is overlaid on the right side. The form includes a search bar at the top, navigation links (Explore, Pricing, Sign In, Sign Up), and a "Get Started Today for Free" section. The sign-up section has fields for Docker ID, Email, and Password, a checkbox for product updates, a CAPTCHA, and a "Sign Up" button. At the bottom, there is a link to the Terms of Service, Privacy Policy, and Data Processing Terms.

dockerhub Search for great content (e.g., mysql) Explore Pricing Sign In Sign Up

Build and Ship any Application Anywhere

Docker Hub is the world's easiest way to create, manage, and deliver your teams' container applications.

Get Started Today for Free

Already have an account? [Sign In](#)

Docker ID

Email

Password

☐ Send me occasional product updates and announcements.

☐ I'm not a robot

[Sign Up](#)

By creating an account, you agree to the [Terms of Service](#), [Privacy Policy](#), and [Data Processing Terms](#).

Connect to the Dockerhub

Create a Dockerhub account

```
docker login
```

Username: Username

Password: Password



Push to the Dockerhub

Tag the docker image with Dockerhub

```
docker tag iris:3.0 subashgandyer/iris:3.0
```

Push the tagged docker image

```
docker push subashgandyer/iris:3.0
```

Google Cloud Platform

Create a new account

console.google.com

The screenshot shows the Google Cloud Platform console interface. At the top, a banner reads: "Your free trial is waiting: activate now to get \$300 credit to explore Google Cloud products. [Learn more](#)". In the top right corner, there are "DISMISS" and "ACTIVATE" buttons. The main header includes the "Google Cloud Platform" logo, a "Select a project" dropdown, and a search bar. The left sidebar contains navigation links: "Cloud Storage", "Browser", "Monitoring", "Settings", and "Release Notes". The main content area is titled "Browser" and contains a "To view this page" message. A "CREATE PROJECT" button is visible on the right. Overlaid on the console is a "Google Cloud Platform" dialog box titled "Welcome AI!". The dialog box contains the following text: "Create and manage your Google Cloud Platform instances, disks, networks, and other resources in one place." Below this is a "Country" dropdown menu with "Canada" selected. Under the "Terms of Service" section, there is a checkbox labeled "I agree to the [Google Cloud Platform Terms of Service](#), and the terms of service of [any applicable services and APIs](#)." Under the "Email updates" section, there is a checkbox labeled "I would like to receive periodic emails on news, product updates and special offers from Google Cloud and Google Cloud Partners." At the bottom right of the dialog box is the "AGREE AND CONTINUE" button.

Google Cloud Platform

Dashboard

 Your free trial is waiting: activate now to get \$300 credit to explore Google Cloud products. [Learn more](#) DISMISS ACTIVATE

 Google Cloud Platform Select a project  Search products and resources     

 Home > Dashboard

 Recent

 Marketplace

 Billing

 APIs & Services >

 Support >

 IAM & Admin >

 Getting started

 Compliance

 Security >

 Anthos >

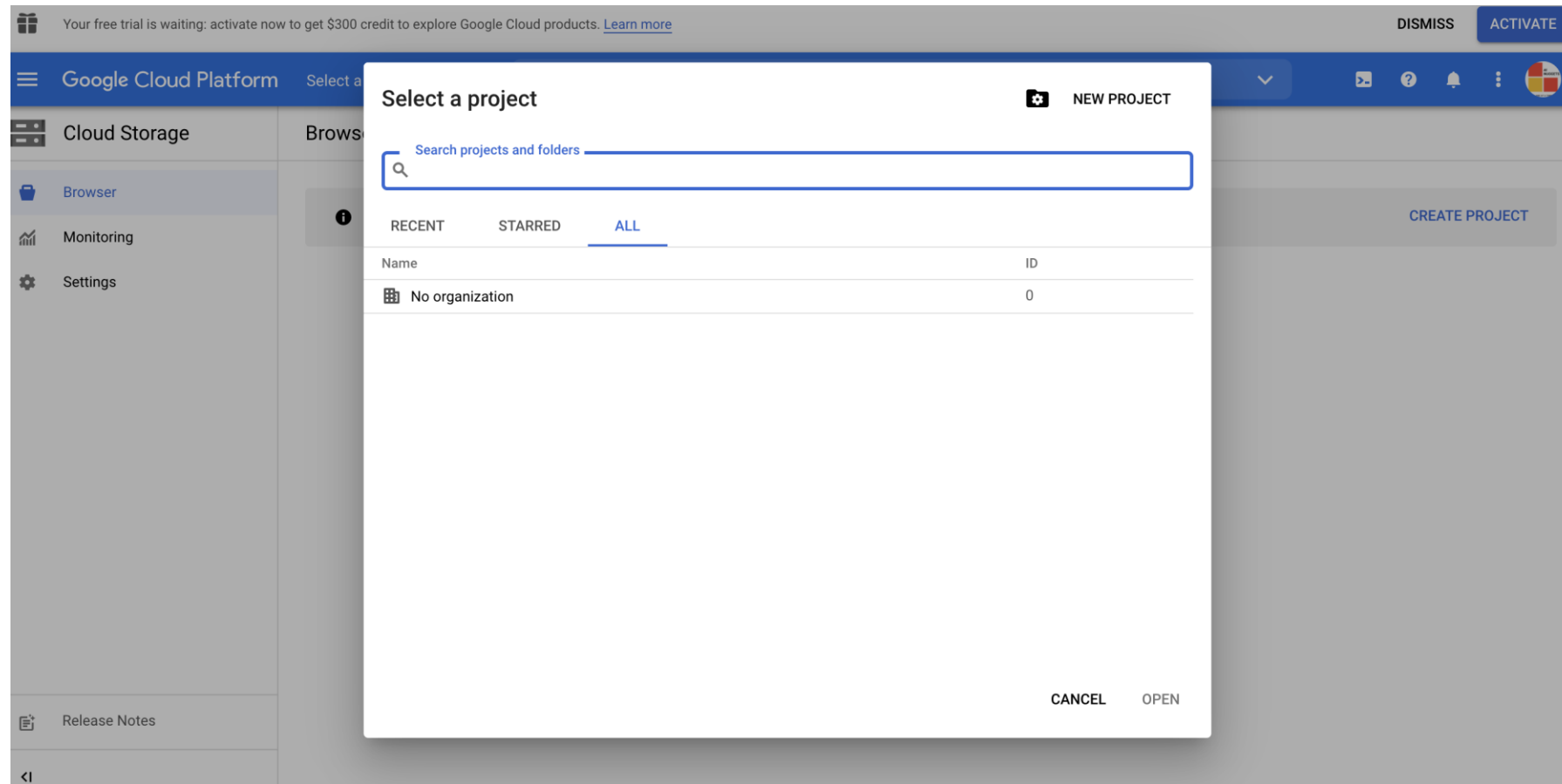
COMPUTE

 Compute Engine >

 To view this page, select a project. [CREATE PROJECT](#)

Google Cloud Platform

Create a new project



Create a new project

☰ Google Cloud Platform

🔍 Search products and resources

New Project



You have 12 projects remaining in your quota. Request an increase or delete projects. [Learn more](#)

[MANAGE QUOTAS](#)

Project name *

GBC-1



Project ID: gbc-1-322717. It cannot be changed later. [EDIT](#)

Location *



No organization

[BROWSE](#)

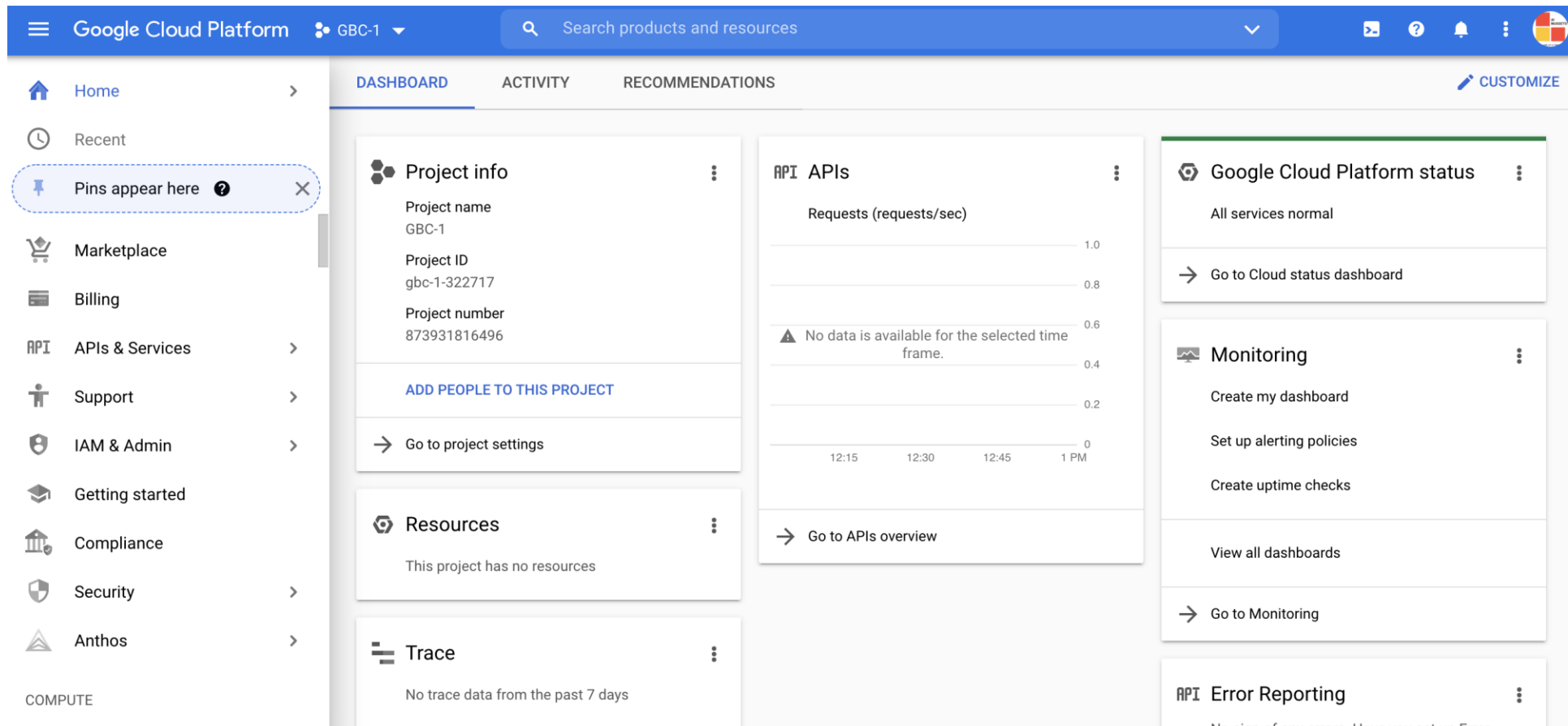
Parent organization or folder

CREATE

CANCEL

New Project

Dashboard



The screenshot shows the Google Cloud Platform (GCP) dashboard for a new project named "GBC-1". The interface is divided into several sections:

- Header:** Includes the Google Cloud Platform logo, the project name "GBC-1", a search bar for products and resources, and navigation icons for home, help, notifications, and settings.
- Left Navigation Panel:** Contains links to Home, Recent, Pins (with a note "Pins appear here"), Marketplace, Billing, APIs & Services, Support, IAM & Admin, Getting started, Compliance, Security, and Anthos. A "COMPUTE" section is also visible at the bottom.
- Dashboard Tabs:** The "DASHBOARD" tab is selected, with other tabs for "ACTIVITY" and "RECOMMENDATIONS". A "CUSTOMIZE" button is located in the top right of the dashboard area.
- Project Info:** Displays the project name "GBC-1", Project ID "gbc-1-322717", and Project number "873931816496". It includes a link to "ADD PEOPLE TO THIS PROJECT" and a button to "Go to project settings".
- APIs:** Shows a line chart for "Requests (requests/sec)" over time. A warning message states: "No data is available for the selected time frame." A button to "Go to APIs overview" is provided.
- Resources:** Indicates that "This project has no resources".
- Trace:** Shows "No trace data from the past 7 days".
- Google Cloud Platform status:** Reports "All services normal" and includes a link to "Go to Cloud status dashboard".
- Monitoring:** Offers options to "Create my dashboard", "Set up alerting policies", and "Create uptime checks". It also includes a link to "View all dashboards" and a button to "Go to Monitoring".
- Error Reporting:** Shows "No sign of any errors. Have you set up Error Reporting?"

Enable Compute Engine API

Google Cloud Platform GBC-1

←

Compute Engine API

Google Enterprise API

Compute Engine API

ENABLE TRY THIS API

OVERVIEW DOCUMENTATION SUPPORT

Overview

Creates and runs virtual machines on Google Cloud Platform.

Additional details

Type: [SaaS & APIs](#)
Last updated: 7/22/21
Category: [Compute](#), [Networking](#), [Google Enterprise APIs](#)
Service name: compute.googleapis.com

Compute Engine API

Google Enterprise API

Compute Engine API

Enable billing for project "GBC-1"

You are not an administrator of any billing accounts. To enable billing on this project, create a new billing account or contact your billing account administrator to enable billing for you. [Learn more](#)

CANCEL CREATE BILLING ACCOUNT

OVERVIEW DOCUMENTATION

Billing Info for Credit Card

 Try Google Cloud for free

Step 1 of 3 Account Information



AI Nuggets

ainuggetsforyou@gmail.com

[SWITCH ACCOUNT](#)

Country

Canada

Terms of Service

☒ I have read and agree to the [Google Cloud Platform Free Trial Terms of Service](#).

Required to continue

CONTINUE

Access to all Cloud Platform Products

Get everything you need to build and run your apps, websites and services, including Firebase and the Google Maps API.

\$300 credit for free

Put Google Cloud to work with \$300 in credit to spend over the next 90 days.

No autocharge after free trial ends

We ask you for your credit card to make sure you are not a robot. You won't be charged unless you manually upgrade to a paid account.



Billing Info for Credit Card

Step 2 of 3 Identity Verification and Contact Information

We'll send a text message with a 6-digit code to verify your identity and confirm where we can reach you about solutions to support your Cloud experience. Standard rates apply.



+1 Phone number

❗ Enter a phone number.

SEND CODE

Access to all Cloud Platform Products

Get everything you need to build and run your apps, websites and services, including Firebase and the Google Maps API.

\$300 credit for free

Put Google Cloud to work with \$300 in credit to spend over the next 90 days.

No autocharge after free trial ends

We ask you for your credit card to make sure you are not a robot. You won't be charged unless you manually upgrade to a paid account.

Billing Info for Credit Card

Step 3 of 3 Payment Information Verification

Your payment information helps us reduce fraud and abuse. You won't be charged unless you turn on automatic billing.

Customer info



Account type ⓘ



Business ▼



Name and address ⓘ

Business name

Billing Info for Credit Card

Payment method ⓘ



Add credit or debit card



Card details



Credit or debit card address is same as above

START MY FREE TRIAL

Create VM instance

Google Cloud Platform

Home

Recent

Pins appear here

Support

IAM & Admin

Getting started

Compliance

Security

Anthos

COMPUTE

Compute Engine

Kubernetes Engine

VMware Engine

VIRTUAL MACHINES

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Committed use discounts

Migrate for Compute Engine

STORAGE

Disks

Snapshots

Images

INSTANCE GROUPS

Instance groups

Health checks

VM MANAGER

OS patch management

OS configuration management

Google Cloud Platform

GBC-1

Search products and resources

Compute Engine

VM instances

CREATE INSTANCE

OPERATIONS

HELP ASSISTANT

HIDE INFO PANEL

Virtual machines

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Committed use discounts

Migrate for Compute Engi...

Storage

Disks

Snapshots

Images

Marketplace

Release Notes

Filter

Status

Name

Zone

Recommendations

Select an instance

PERMISSIONS

LABELS

MONITORING

Please select at least one resource.

VM Instances

Compute Engine lets you use virtual machines that run on Google's infrastructure. Create micro-VMs or larger instances running Debian, Windows, or other standard images. Create your first VM instance, import it using a migration service, or try the quickstart to build a sample app.

CREATE INSTANCE

TAKE THE QUICKSTART

Create VM instance

[← Create an instance](#)

To create a VM instance, select one of the options:

New VM instance

Create a single VM instance from scratch

New VM instance from template

Create a single VM instance from an existing template

New VM instance from machine image

Create a single VM instance from an existing machine image

Marketplace

Deploy a ready-to-go solution onto a VM instance

Name ?
Name is permanent

Labels ? (Optional)

+ Add label

Region ?
Region is permanent

us-central1 (Iowa)

Zone ?
Zone is permanent

us-central1-a

Machine configuration

Machine family

General-purpose

Compute-optimized

Memory-optimized

GPU

Machine types for common workloads, optimized for cost and flexibility

Series

E2

CPU platform selection based on availability

Machine type

e2-medium (2 vCPU, 4 GB memory)



vCPU

1 shared core

Memory

4 GB

GPUs

-

< CPU platform and GPU

Confidential VM service ?
☐ Enable the Confidential Computing service on this VM instance.

You have \$377.66 free trial credits

\$25.46 monthly estimate

That's about \$0.035 hourly

Pay for what you use: No upfront costs

< Details

Create VM instance

Boot disk ?



New 10 GB balanced persistent disk

Image

 Debian GNU/Linux 10 (buster)

Change

Identity and API access ?

Service account ?

Compute Engine default service account

Access scopes ?

☐ Allow default access

☒ Allow full access to all Cloud APIs

☐ Set access for each API

Firewall ?

Add tags and firewall rules to allow specific network traffic from the Internet

- ☒ Allow HTTP traffic
- ☒ Allow HTTPS traffic

Management, security, disks, networking, sole tenancy

Your free trial credit will be used for this VM instance. [GCP Free Tier](#)

CreateCancel

Allow full access to all Cloud APIs

Allow HTTP traffic

Allow HTTPS traffic

VM instance Created

34.136.32.211 [IP address]

Yours will be different

VM instances

 CREATE INSTANCE

 IMPORT VM



 OPERATIONS ▾

 HELP ASSISTANT

 LEARN

SHOW INFO PANEL

INSTANCES

INSTANCE SCHEDULE

VM instances are highly configurable virtual machines for running workloads on Google infrastructure. [Learn more](#)

 Filter Enter property name or value



☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	✓	demo-1	us-central1-a			10.128.0.2 (nic0)	34.136.32.211 	SSH ▾ 

Create a Firewall rule

Home

Recent

Pins appear here ?

Cloud Scheduler

Cloud Tasks

Workflows

NETWORKING

VPC network

Network services

Hybrid Connectivity

Network Service Tiers

Network Security

Network Intelligence

OPERATIONS

Monitoring

M instances

INSTANCES

M instances are highly confi

frastructure. [Learn more](#)

Filter Enter property

Status

related actions

VPC networks

External IP addresses

Bring your own IP

Firewall

Routes

VPC network peering

Shared VPC

Serverless VPC access

Packet mirroring

Firewall

+ CREATE FIREWALL RULE

REFRESH

CONFIGURE LOGS

DELETE

VPC network

VPC networks

External IP addresses

Bring your own IP

Firewall

Routes

VPC network peering

Shared VPC

Serverless VPC access

Packet mirroring

Create a firewall rule

Firewall rules control incoming or outgoing traffic to an instance. By default, incoming traffic from outside your network is blocked. [Learn more](#)

Name *

Lowercase letters, numbers, hyphens allowed

Description

Logs

Turning on firewall logs can generate a large number of logs which can increase costs in Cloud Logging. [Learn more](#)

☐ On
 ☒ Off

Network *

default

Priority *

1000

CHECK PRIORITY OF OTHER FIREWALL RULES

Priority can be 0 - 65535

Direction of traffic ?

☒ Ingress
 ☐ Egress

Action on match ?

Create a Firewall rule

Targets

Specified target tags

Target tags *

demo-1

Source filter

IP ranges

Source IP ranges *

0.0.0.0 for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter

None

Protocols and ports

Allow all

Specified protocols and ports

DISABLE RULE

CREATE

CANCEL

Target tags --> demo-1

Source IP ranges --> 0.0.0.0
0.0.0.0/0

Protocols --> Allow all

rules										
☐	demo-1	Ingress	demo-1	IP ranges: 0.0.0.0/0	all	Allow	1000	default	Off	–
☐	default	Ingress	Apply to all	IP ranges: 0.0.0.0/0	all	Allow	65534	default	Off	–

Edit VM instance

← VM instance details

EDIT

RESET

+

■

||

demo-1

Details

Observability

NEW

Screenshot

Remote access

SSH

Connect to serial console

☐

Enable connecting to serial ports ?

Logs

Cloud Logging

Serial port 1 (console)

More

Instance Id

6445889128666555002

Machine type

e2-medium (2 vCPUs, 4 GB memory)

Reservation

Automatically choose

CPU platform

Intel Haswell

Display device

Turn on a display device if you want to use screen capturing and recording tools.

☐

Turn on display device

Zone

us-central1-a

Labels

Network tags

http-server

https-server

demo-1

Network tags --> demo-1

Successfully created firewall rule "demo-1".



SSH

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	✓	demo-1	us-central1-a			10.128.0.2 (nic0)	34.136.32.211 ↗	SSH ⌵ ⋮

Related actions

View billing report
 View and manage your Compute Engine billing

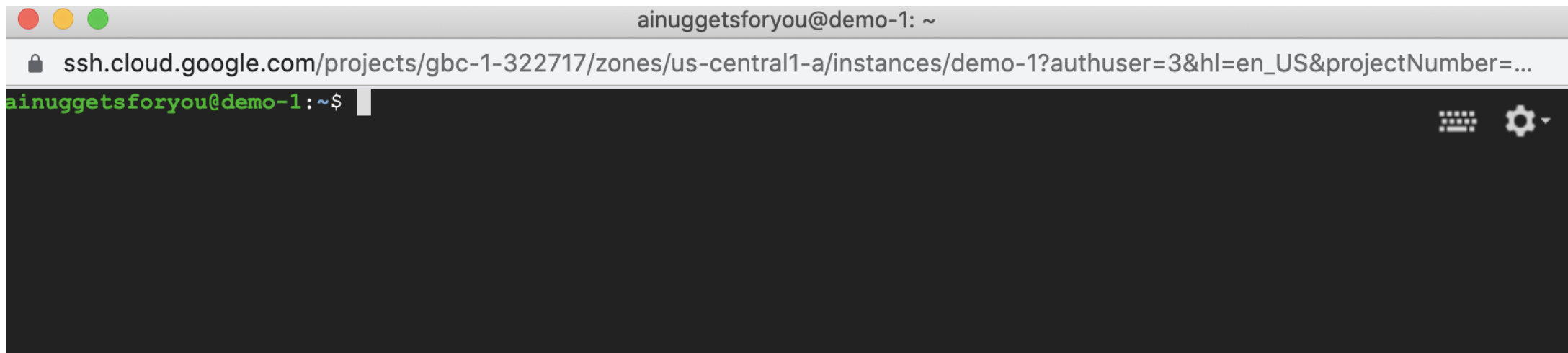
Monitor VMs
 View outlier VMs across metrics like CPU and network

Explore VMs
 View, search, download VM

Patch management
 Schedule patch updates and view patch compliance on VM instances

Open in browser window
 Open in browser window on custom port
 Open in browser window using provided private SSH key
 View gcloud command
 Use another SSH client

DISMISS



Installing Docker

```
sudo apt update
```

```
sudo apt install apt-transport-https ca-certificates curl gnupg2 software-properties-common
```

```
curl -fsSL https://download.docker.com/linux/debian/gpg | sudo apt-key add -
```

```
sudo add-apt-repository "deb [arch=amd64]  
https://download.docker.com/linux/debian $(lsb_release -cs) stable"
```

```
sudo apt update
```

```
apt-cache policy docker-ce
```

```
sudo apt install docker-ce
```

```
sudo systemctl status docker
```

Pulling a docker image

Pulling a Docker image

```
sudo docker pull subashgandyer/iris:3.0
```

```
ainuggetsforyou@demo-1:~$ sudo docker pull subashgandyer/iris:3.0
3.0: Pulling from subashgandyer/iris
ac2522cc7269: Pull complete
8edbc159ce36: Pull complete
0e6f0a09ecb9: Pull complete
cb3ef6c5b062: Pull complete
10b895c4ca73: Pull complete
5a3b312dc2c0: Pull complete
379c7f8f85be: Pull complete
Digest: sha256:3d6764f710601a685074744150ad3bc4ecc30fd17093439b3beaf938a12f05eb
Status: Downloaded newer image for subashgandyer/iris:3.0
docker.io/subashgandyer/iris:3.0
```

View docker image

Viewing Docker images in remote GCP machine

```
sudo docker images
```

```
ainuggetsforyou@demo-1:~$ sudo docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
subashgandyer/iris	3.0	9eaa93182d41	18 hours ago	524MB

Running a docker container

Running a Docker container inside GCP VM instance

```
sudo docker run -d --restart on-failure -p  
5000:5000 subashgandyer/iris:3.0
```


View docker containers

Viewing Docker containers in remote GCP machine

```
sudo docker ps
```

```
ainuggetsforyou@demo-1:~$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
	NAMES				
c324a89e89f7	subashgandyer/iris:3.0	"python app.py"	22 seconds ago	Up 20 seconds	0.0.0.0:5000->5000/tcp,
:::5000->5000/tcp	dazzling_pascal				

Testing the docker container endpoints

Curl

POSTMAN

Python Web Client in browser



Curl Calls

```
curl http://34.136.32.211:5000/train
```

```
curl http://34.136.32.211:5000/
```

```
curl -X POST -d  
'sepalLength=1.5&sepalWidth=3.5&petalLength=4.5&petalWidth=5.6'  
http://34.136.32.211:5000/final\_classifier
```

POSTMAN – train a model

▶ Train Examples 0 ▾ BUILD

GET ▾ http://34.136.32.211:5000/train Send ▾

Params Authorization Headers (7) Body Pre-request Script Tests Settings C

Query Params

KEY	VALUE	DESCRIPTION	...
Key	Value	Description	

Body Cookies Headers (4) Test Results 🌐 Status: 200 OK Time: 130 ms Size: 198 B Save

Pretty Raw Preview Visualize JSON ▾ 🔄

```
1 {  
2   "code": 200,  
3   "message": "Model is created."  
4 }
```



POSTMAN – classify a sample

► Predict Examples 0 ▾ BUILD

POST ▾ http://34.136.32.211:5000/final_classifier Send ▾

Params ● Authorization Headers (9) **Body ●** Pre-request Script Tests Settings

● none ● form-data ● x-www-form-urlencoded ● raw ● binary ● GraphQL

	KEY	VALUE	DESCRIPTION	...
✓	sepalLength	1.3		
✓	sepalWidth	2.4		
✓	petalLength	5.5		
✓	petalWidth	5.4		
	Key	Value	Description	

Body Body Cookies Headers (4) Test Results 🌐 Status: 200 OK Time: 117 ms Size: 251 B Save

Pretty Raw Preview Visualize HTML ▾ 

```
1 <html>
2 <h1>IRIS Classifier Results</h1>
3 <h3> Predicted Class: Iris-virginica </h3>
4
5 </html>
```

Test with Python Web Client

Navigate to

```
http://34.136.32.211:5000/train
```

Then,

```
http://34.136.32.211:5000/
```

Input the values within all fields

Click Classify button

Further Reading

Docker Tutorial

<https://www.docker.com>

Google Cloud Platform

<https://cloud.google.com/docs>

